SUNWAY UNIVERSITY · SCHOOL OF MATHEMATICAL SCIENCES & COMPUTING

Autonomous Wildfire Risk Assessment & UAV Surveillance

Mamdani fuzzy inference driving a four-probe UAV inside an NVIDIA Isaac Sim digital twin

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BACKGROUND

Why a threshold is the wrong tool

THE PROBLEM

A single threshold cannot see interaction.

A 40 °C cut-off cannot separate 38 °C at 15 % humidity and 70 km/h wind from 42 °C in saturated still air. The first case is the dangerous one.

Spread is non-linear.

Dry wind at moderate temperature ignites faster than a hot, humid afternoon, so a temperature-only trigger fires late.

The telemetry is noisy.

Rotor vibration and turbulence corrupt raw readings, so a hard edge flickers between alarm states.

THE APPROACH

Continuous membership.

Four overlapping sets per variable replace the cliff edge, so the output moves smoothly with the input.

Four-probe fusion.

Temperature, relative humidity, wind speed and fuel dryness combine into one fire risk index.

The output flies the aircraft.

Throttle scales from 20 % patrol to 90 % emergency, so the UAV closes its own loop above a hotspot.

SYSTEM PIPELINE

Sensing to actuation in five stages

01

Synthetic sensing

Four probes on the OpenUSD stage, sampled at 60 Hz.

02

Fuzzification

16 input sets, four per variable, triangular and trapezoidal.

03

Fuzzy inference

10 expert rules; min for AND, max for aggregation.

04

Defuzzification

Centroid of area; continuous output, 0.28 ms median per loop.

05

Actuation and alert

Throttle 20-90 %; alert 0-3 as a third fuzzy output.

60 Hz

sensor tick from the Isaac Sim stage

0.28 ms

median per loop, p99 1.42 ms

0.6 %

of the 50 ms control budget

FUZZY ENGINE

Linguistic variables and membership functions

INPUTS

Temperature (T)
0 - 50 °C low · moderate · high · extreme

Relative humidity (RH)
0 - 100 % very low · low · moderate · high

Wind speed (WS)
0 - 100 km/h calm · moderate · high · severe

Fuel dryness (FD)
0 - 100 % wet · moderate · dry · critical

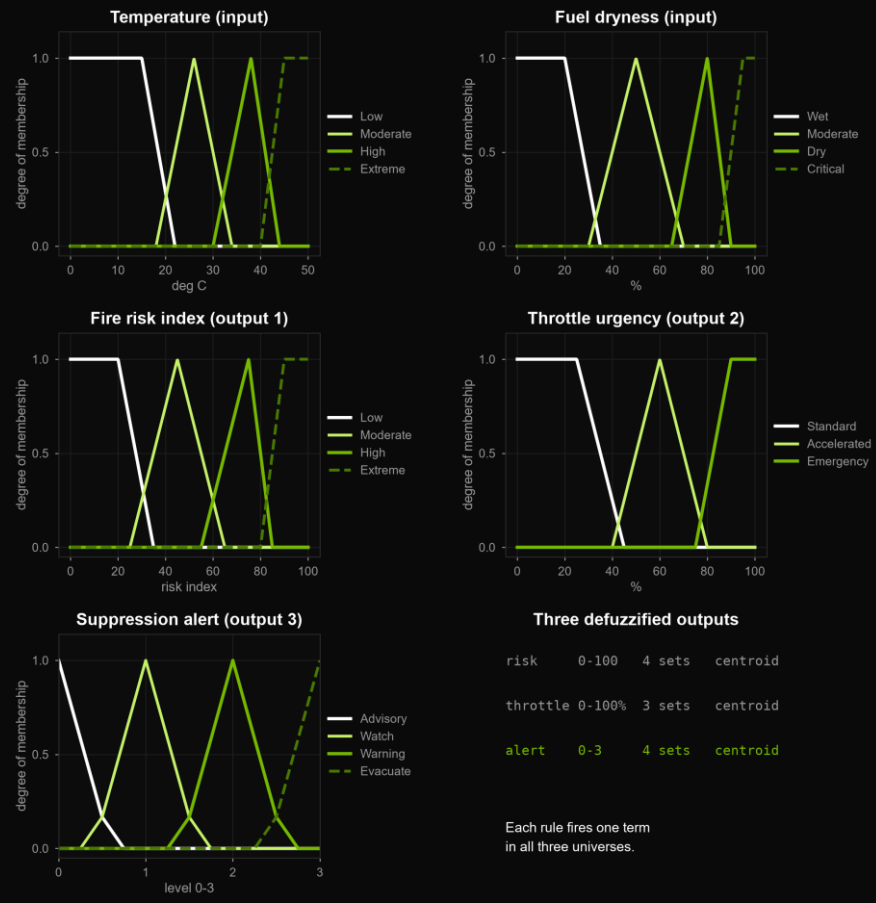
OUTPUTS

Fire risk index (FRI)
0 - 100 low · moderate · high · extreme

Throttle urgency
0 - 100 % standard · accelerated · emergency

Suppression alert
0 - 3 advisory · watch · warning · evacuate

Trapezoids close the outermost sets, so readings past the sensor range saturate instead of running off the universe. Each rule fires one term in all three universes.



Temperature and fuel dryness shown; humidity and wind use the same four-set template. Output 3 adds the alert consequent.

KNOWLEDGE BASE

Ten curated rules out of 256 combinations

RULE	ANTECEDENT	FIRE RISK	THROTTLE	ALERT (0-3)
R09 · R07	Temperature extreme AND (fuel critical OR humidity very low)	92	90 %	2.87
R04	Temperature moderate AND wind high AND fuel dry	71	60 %	2.00
R03	Temperature moderate AND fuel moderate	45	20 %	1.00
R01 · R10	Humidity high AND (fuel wet OR wind calm)	20	20 %	1.00

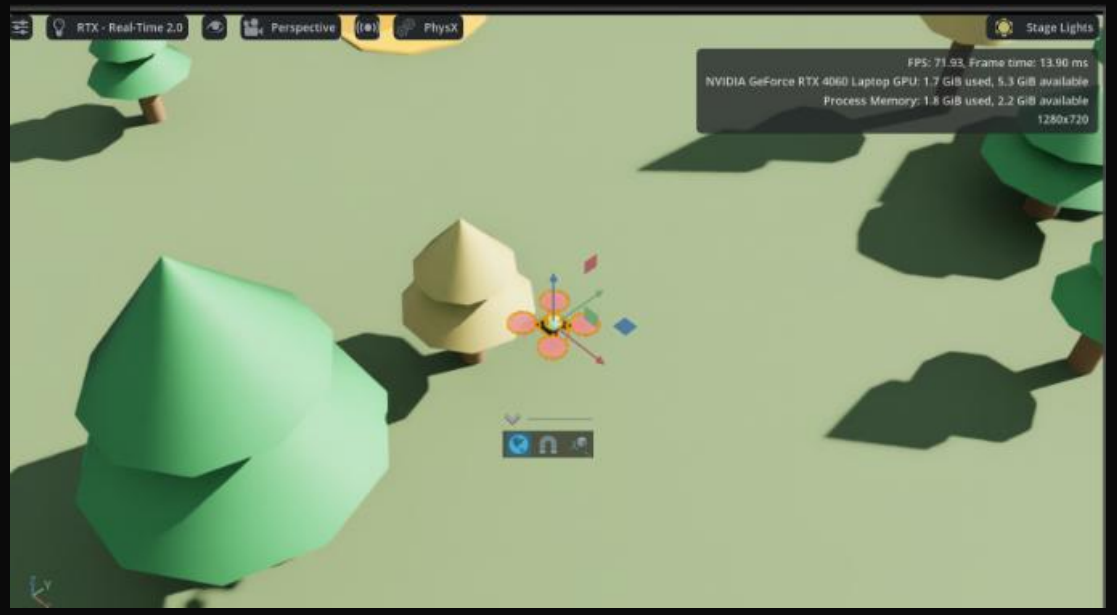
4 inputs × 4 sets = 256 antecedent combinations, curated down to the 10 that are physically meaningful.
Mamdani: min for AND, max for aggregation, centroid for defuzzification.
Cost: 0.28 ms per loop on the workstation; each rule fires all three consequents.

3D DIGITAL TWIN

NVIDIA Isaac Sim 6.2.0 — procedural forest stage



65 m procedural forest: 48 pines, hotspot plumes, chase camera



UAV prim over hotspot 01 — probe readings feed the next inference tick

65 × 65 m

OpenUSD stage, procedurally generated

3

active hotspots, up to 44.5 °C core plumes

2

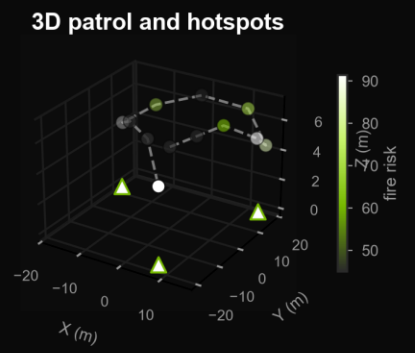
cameras: chase view and thermal gimbal

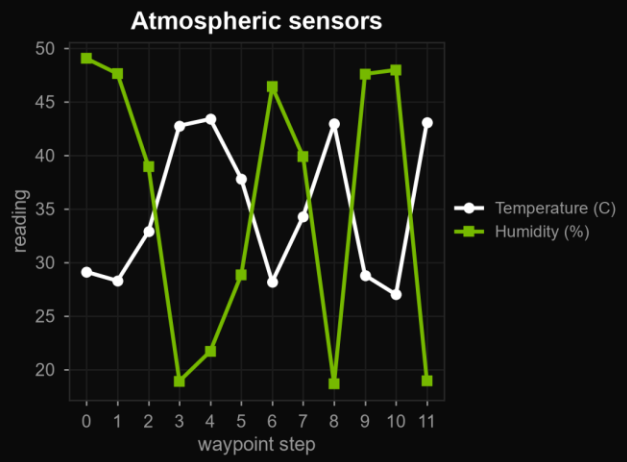
67-72

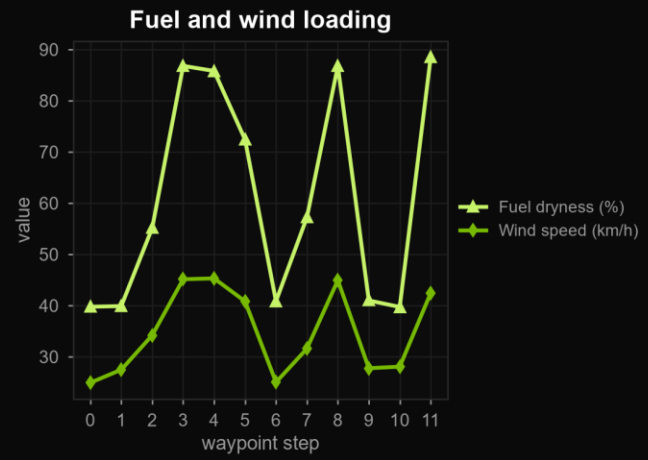
FPS in RTX Real-Time 2.0 at 1280 × 720

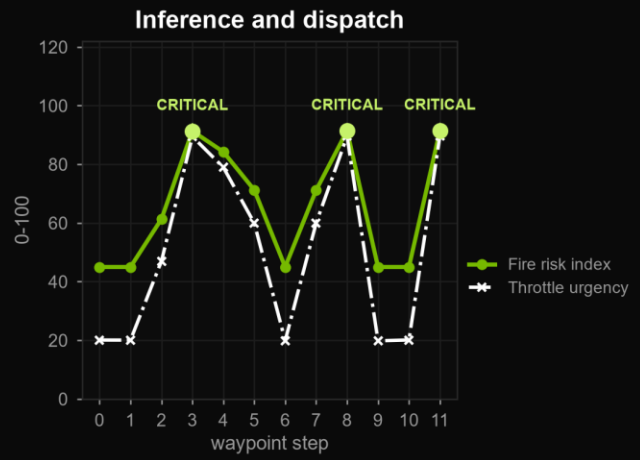
FLIGHT RESULTS

Twelve waypoints, closed loop, no retuning









WHAT THE RUN SHOWS

Baseline, WP 00-01

29.1 °C, RH 49.0 %. FRI 45.0 holds at WATCH, throttle 20.1 %, alert 1.000.

Hotspot entry, WP 03

42.8 °C, RH 18.9 %, fuel 86.8 %. FRI 91.4, throttle 89.5 %, alert 2.826.

Second plume, WP 08

42.9 °C, RH 18.7 %. FRI 91.4, alert 2.829 — same response, no retuning.

Recovery, WP 06

28.2 °C, RH 46.4 %. FRI returns to 45.0, throttle 19.9 %, alert 1.000.

Numbers are the printed output of run_stage_patrol_simulation(seed=0), reproducible from the repository.

DESIGN DECISION

Why centroid, when all three are this cheap

Method	Critical FRI	Actuator smoothness	Noise	Flight suitability
Centroid (COG)	92.2	continuous	high	**selected**
Bisector	92.5	good	moderate	viable
Mean of max (MOM)	95.1	step jumps	low	unsafe

Continuity, not speed.

Centroid integrates the whole aggregated area, so a small input change produces a small output change. On a 250 g airframe that is the difference between smooth thrust and torque steps.

MOM jumps between peaks.

Mean of maxima returns the average of the aggregated set's maxima, so as the input crosses a boundary the output leaps to the next peak: 95.1 against 92.2.

All three are cheap.

A full evaluation loop is 0.28 ms, so none of the methods is the bottleneck. They differ only in the 2.9-point gap MOM shows at the critical scenario.

Bisector is the compromise.

Smooth and within 0.3 points of centroid on every scenario, but no better on the flight logs, so the simpler method wins.

FEASIBILITY

It fits on the aircraft

EDGE STACK

Jetson Orin Nano, 8 GB.
7-15 W envelope, inside a micro-UAV battery budget.

0.28 ms inference.
median per loop; 0.6 % of the 50 ms budget.

Sensors already proven.
FLIR Lepton 3.5 LWIR for the plume, Sensirion SHT40 for temperature and humidity.

LoRa backhaul.
Carries the GPS fix of a confirmed hotspot to the ground crew.

WHY IT MATTERS

SDG 13.1, climate action.
Early thermal detection catches a fire before crown spread, which is where the emission saving is.

SDG 15.2, life on land.
Protects forest reserves and the habitat inside them.

Operational value.
Exact coordinates cut containment time, water use and crew exposure.

Offline by design.
The decision survives link loss, which is exactly when the fire matters most.

CONCLUSION

Delivered, and what is next

DELIVERED

Fuzzy engine.
Four inputs, ten rules, verified against the report's test vectors.

Digital twin.
65 m procedural forest holding 67-72 FPS in real-time ray tracing.

Closed-loop actuation.
20 % to 90 % throttle demonstrated over a live hotspot.

Edge speed.
0.28 ms median per loop, p99 1.42 ms; edge figure projected, not measured.

NEXT

ANFIS.
Replace hand-tuned memberships with functions fitted to satellite burn history.

Multi-UAV mesh.
Decentralised coverage for areas a single airframe cannot hold.

Field trials.
PX4 and Matrice 300 under a controlled-burn permit.

Limitation, stated plainly: every result here is simulation-only, and the rule base is expert-tuned rather than learned.

Thank you

Questions, then the live Isaac Sim run

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github.com/123haiku/csc3034-fuzzy-wildfire-sim

Slides, figures and simulation output regenerate from the repository.